

TryHackMe — Pre-Security Learning Path

Extending Your Network

Learning Notes — Port Forwarding, Firewalls, VPNs & LAN Devices

"A network alone is not enough — the technologies that extend, protect, and connect networks are what make modern computing possible."

 Port Forwarding Firewalls VPN Technologies LAN Devices

1. Introduction to Port Forwarding

Port forwarding (also called port mapping) is an essential component in connecting applications and services to the Internet. Without it, services like web servers, game servers, and remote desktop are only accessible to devices within the same local network — a configuration known as an intranet. Port forwarding bridges the gap between private internal networks and the public Internet.

The Core Problem Port Forwarding Solves

By default, a server on IP 192.168.1.10 port 80 is invisible to the outside world. Only devices on the same LAN can reach it. Port forwarding maps a public-facing port on the router to that internal server — making it reachable from anywhere on the Internet via the router's public IP address.

1.1 How Port Forwarding Works

Port forwarding is configured at the router. The router maintains a table of port-forwarding rules that map incoming traffic on a specific public port to a specific internal device and port. When traffic arrives at the router matching a rule, it is redirected (forwarded) to the correct internal device.

Step	What Happens	Example
1. Without forwarding	A request from the Internet to access a web server on Network #1 is dropped — the router has no rule to handle it.	External user → Router (82.62.51.70:80) → No rule → Request dropped
2. Rule added	Admin adds a port-forwarding rule: incoming traffic on public port 80 → forward to 192.168.1.10:80 internally.	Admin configures: public 80 → 192.168.1.10:80
3. With forwarding	Now, Network #2 can reach the web server on Network #1 using Network #1's public IP address.	External user → Router (82.62.51.70:80) → 192.168.1.10:80 ✓

1.2 Port Forwarding vs. Firewalls — Key Distinction

Port forwarding and firewalls are often confused because both deal with ports. They serve entirely different purposes and work at different stages of traffic control:

Aspect	Port Forwarding	Firewall
What it does	Opens specific ports — maps external port to an internal device	Controls whether traffic is allowed to travel through open ports
Where configured	At the router	At the firewall device (standalone or embedded in router/OS)
Direction	Inbound — external requests to internal services	Both inbound and outbound traffic filtering
Analogy	Unlocking a door so someone can knock on it	Deciding whether to answer the door based on who is knocking
Security role	Not a security feature — creates exposure that must be managed	The primary network security enforcement mechanism

Security Warning

Every port-forwarding rule is a potential attack vector. Any service exposed to the Internet must be hardened, kept patched, and monitored. Attackers routinely scan public IPs for open ports — anything that responds is a target. Unnecessary port-forwarding rules should always be removed.

2. Firewalls 101

A firewall is a device (hardware or software) within a network responsible for determining what traffic is permitted to enter and exit. It acts as a gatekeeper — inspecting traffic against a set of rules and either allowing or dropping it based on those rules.

OSI Layers of Operation

Firewalls operate at Layer 3 (Network) and Layer 4 (Transport) of the OSI model — inspecting IP addresses, ports, and protocols. Next-generation firewalls (NGFWs) extend this to Layer 7 (Application), enabling inspection of actual application content.

2.1 The Two Primary Firewall Categories

Category	Stateful	Stateless
How it works	Uses the ENTIRE information from a connection. Tracks connection state and context to determine behaviour.	Uses a STATIC set of rules. Inspects each individual packet in isolation — no memory of previous packets.
Intelligence	High — understands whether a packet is part of an established, expected connection.	Low — each packet is evaluated against fixed rules only. No context awareness.
Resource usage	Higher — must maintain connection state tables for all tracked sessions.	Lower — simple rule matching requires minimal processing and memory.
Adaptability	Adaptive — can detect anomalies within a connection (e.g. unexpected RST floods).	Not adaptive — a packet matching an ALLOW rule is always allowed, regardless of context.
Best use	Protecting networks where sophisticated, context-aware filtering is needed.	High-throughput environments where speed matters more than intelligence (e.g. backbone filtering).
Attack detection	Can detect port scans, session hijacking attempts, and abnormal connection behaviour.	Cannot — a series of suspicious packets matching individual allow rules will pass through freely.

2.2 Firewall Rules — How They Work

Firewalls enforce policy through a set of rules (also called ACLs — Access Control Lists). Each rule defines: the source IP/port, destination IP/port, protocol, and the action to take (ALLOW or DROP/DENY). Rules are evaluated in order — the first matching rule wins.

Action	Source IP	Src Port	Dest IP	Dest Port	Protocol
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Action	Source IP	Src Port	Dest IP	Dest Port	Protocol
ALLOW	192.168.1.0/24	ANY	203.0.110.1	80	HTTP
ALLOW	192.168.1.0/24	ANY	203.0.110.1	443	HTTPS
DROP	198.51.100.34	ANY	203.0.110.1	80	HTTP
ALLOW	ANY	ANY	ANY	22	SSH
DROP	ANY	ANY	ANY	ANY	ANY

Rule Order Matters — Default Deny

Firewall rules are processed top-to-bottom. The last rule in any good firewall policy is "DROP ANY ANY" — denying all traffic that does not match an explicit ALLOW rule above it. This is called default-deny and is a security best practice. Without it, any unmatched traffic would be allowed through.

2.3 Hands-On Lab — Firewall DoS Defence (THM{FIREWALLS_RULE})

The TryHackMe room includes a practical firewall simulation where your server is being flooded with packets from an attacker — a Denial of Service (DoS) attack. Your task is to configure a firewall rule to block the attacker's traffic before the server crashes.

Step	Action	Detail
1	Observe the simulation	Two computers send packets to the server (203.0.110.1). One is a legitimate user; one is the attacker flooding the server with traffic.
2	Identify the attacker	The attacker's IP address is 198.51.100.34 — visible from the high volume of packets it is sending compared to the legitimate user.
3	Add a DROP rule	In the Firewall Rules panel, set: Source IP = 198.51.100.34, Destination IP = 203.0.110.1, Port = 80, Protocol = HTTP, Action = DROP.
4	Apply the rule	The firewall drops all packets from the attacker's IP — traffic from the legitimate user continues flowing through unaffected.
5	Server recovers	With the flood stopped, the server's load meter returns to normal. The flag is revealed: THM{FIREWALLS_RULE}.

Flag Captured — THM{FIREWALLS_RULE}

This lab demonstrates a real-world DoS scenario. The attacker at 198.51.100.34 floods port 80 on the target server 203.0.110.1. A single DROP rule targeting that source IP is enough

to stop the attack — showing how firewalls are the primary tool for blocking malicious traffic at the network perimeter.

3. VPN — Virtual Private Network

A Virtual Private Network (VPN) is a technology that allows devices on separate, geographically dispersed networks to communicate securely as if they were on the same private network. It achieves this by creating an encrypted tunnel through the public Internet — making remote devices appear locally connected.

The VPN Tunnel Analogy

Imagine two offices in different cities. Without a VPN, employees at each office can only access their own local resources. With a VPN, both offices create an encrypted tunnel through the Internet — they can now communicate as if they were sitting in the same building, even though they are hundreds of miles apart.

3.1 Benefits of VPNs

Benefit	What It Means
Connects remote networks	Two or more offices, data centres, or remote workers can be connected as if on the same local network — regardless of physical location.
Privacy from ISP	Internet traffic is encrypted inside the VPN tunnel. The ISP (Internet Service Provider) can see that a VPN connection exists but cannot read the content.
Anonymity online	The destination website sees the VPN server's IP address, not the user's real IP — protecting identity and location.
Protects on public Wi-Fi	On untrusted networks (cafes, airports), the VPN encrypts all traffic — preventing eavesdropping and man-in-the-middle attacks.
Bypasses geo-restrictions	Traffic appears to originate from the VPN server's location — enabling access to region-restricted content and services.
Secure remote access	Employees working from home can securely access internal company resources as if physically present in the office.
TryHackMe uses VPN	THM's vulnerable machines are not exposed to the open Internet — you connect via VPN to access them safely in an isolated environment.

3.2 VPN Technologies

Three VPN technologies are covered in the room. Understanding their differences — particularly their encryption and routing capabilities — is important for both implementation and security assessment:

Technology	Full Name	Description
PPP	Point-to-Point Protocol	The foundational VPN building block. Provides authentication and basic encryption of data between two direct endpoints. Non-routable — PPP traffic cannot leave a network by itself. Used by PPTP as its underlying encryption and authentication mechanism.
PPTP	Point-to-Point Tunnelling Protocol	Wraps PPP in a tunnel, enabling it to traverse networks (solving PPP's non-routable limitation). Very easy to set up and widely supported — but weakly encrypted compared to modern alternatives. Largely deprecated for security-sensitive use cases due to known cryptographic vulnerabilities.
IPSec	Internet Protocol Security	Encrypts data using the existing Internet Protocol (IP) framework — operating at Layer 3. Strong encryption with robust authentication. More complex to configure than PPTP. Widely used for site-to-site VPNs and remote access solutions where security cannot be compromised.

PPP vs IPSec — The Key Difference

PPP only encrypts and authenticates the data — it cannot route traffic across networks on its own (non-routable). IPSec uses the IP framework to both encrypt data AND route it across networks. For modern VPNs requiring strong security and cross-network routing, IPSec is the standard choice. PPTP is avoided in security-conscious environments due to weak encryption.

4. LAN Networking Devices

Two hardware devices are essential to understand when building and extending networks: routers and switches. They operate at different OSI layers and serve fundamentally different purposes — routers connect networks together, while switches connect devices within a network.

Router | OSI Layer 3 — Network

A router's job is to connect networks and pass data between them. It does this by routing — finding the optimal path for data to travel from source to destination across multiple networks.

- Operates at Layer 3 (Network Layer) of the OSI model — works with IP addresses, not MAC addresses.
- Connects two or more separate networks together — e.g. your home LAN to the Internet.
- Maintains a routing table — a map of known networks and the best paths to reach them.
- Routing protocols (OSPF, RIP, BGP) allow routers to dynamically discover and share path information.
- When multiple paths exist to a destination, the router selects the most optimal route based on hop count, bandwidth, or reliability.
- Most home/office routers also include integrated firewall and port-forwarding functionality via an admin web interface.
- The router is the device configured for port forwarding — all external traffic enters through the router first.

Security Note: Routers are high-value targets for attackers. Compromising a router gives control over all traffic passing through it — enabling traffic interception, redirection, and man-in-the-middle attacks. Router admin interfaces must be secured with strong passwords, updated firmware, and no default credentials.

Switch | OSI Layer 2 and/or Layer 3

A switch is a dedicated networking device responsible for connecting multiple devices within the same network together. Unlike a router (which routes between networks), a switch facilitates communication between devices on the same LAN.

- Connects multiple devices (computers, servers, printers) within the same LAN using Ethernet cables.
- Can connect between 3 and 63 devices efficiently using standard Ethernet.
- Layer 2 Switch — operates using MAC addresses. Forwards frames only to the port where the destination MAC address is connected. More efficient than a hub (which floods all ports).
- Layer 3 Switch — can also route packets using IP addresses (like a router). Combines switching and basic routing in one device.
- Maintains a MAC address table (CAM table) — maps MAC addresses to physical ports for efficient frame delivery.
- VLANs (Virtual LANs) — Layer 3 switches support VLANs, which segment a physical LAN into multiple logical networks without requiring separate physical switches.

Security Note: ARP poisoning targets Layer 2 switches by flooding the CAM table with fake MAC-to-IP mappings, redirecting traffic through the attacker. VLAN hopping exploits misconfigured trunking to jump between VLANs. Port security features on managed switches can mitigate both attacks.

4.1 Router vs. Switch — Key Comparison

Feature	Router	Switch
OSI Layer	Layer 3 — Network	Layer 2 (standard) or Layer 3 (advanced)
Addressing used	IP addresses	MAC addresses (L2) / IP addresses (L3)
Primary job	Connect separate networks together	Connect devices within the same network
Scope	Inter-network (WAN / Internet connections)	Intra-network (LAN connections)
Traffic handling	Routes packets based on IP destination	Forwards frames based on MAC address destination
Routing table	Yes — maintains paths to remote networks	No (L2) / Yes (L3 switches only)
VLAN support	Not natively (requires subinterfaces)	Yes — L3 switches natively support VLANs
Admin interface	Web UI/console for port forwarding, firewall	Web UI/console for VLAN config, port security

5. Hands-On Lab — Network Simulator

The final task in the room is a network simulator where you must send a TCP packet from Computer 1 to Computer 3 through the network. The lab demonstrates how packets are routed through switches and routers in a real LAN environment.

5.1 Lab Setup

- Computer 1 — the source device initiating the TCP connection.
- Computer 2 — another device on the network (not the target).
- Computer 3 — the destination device receiving the TCP packet.
- Router — routes packets between network segments.
- Switch — connects all LAN devices, forwarding frames based on MAC addresses.

5.2 Steps to Complete the Simulator

Step	Action	What Happens in the Network
1	Select Computer 1	Computer 1 is chosen as the source of the TCP packet.
2	Select TCP protocol	TCP is chosen as the packet type — this uses the reliable connection-oriented transport protocol.
3	Select Computer 3 as target	The destination is set to Computer 3 on the network.
4	Send the packet	The packet travels: Computer 1 → Switch → Router (if cross-subnet) → Switch → Computer 3.
5	TCP handshake occurs	The simulator logs the full TCP Three-Way Handshake — SYN, SYN/ACK, ACK — between the two computers. There are 5 HANDSHAKE entries in the network log.
6	Flag revealed	Once Computer 3 successfully receives the TCP packet, the flag THM{YOU'VE_GOT_DATA} is displayed.

Flags Captured

THM{FIREWALLS_RULE} (Firewall Lab) and THM{YOU'VE_GOT_DATA} (Network Simulator). The network simulator also shows 5 HANDSHAKE log entries, confirming the TCP Three-Way Handshake was successfully established and completed between Computer 1 and Computer 3.

6. All Topics at a Glance — Reference Summary

Topic	One-Line Summary
Port Forwarding	Maps an external public port on a router to an internal device, making services reachable from the Internet.
Intranet	A private, internal network — services accessible only to devices on the same LAN (no Internet exposure).
Firewall	A device that filters network traffic — allows or drops packets based on rules (OSI Layers 3 and 4).
Stateful Firewall	Inspects the ENTIRE connection context — adaptive, intelligent, higher resource usage.
Stateless Firewall	Inspects INDIVIDUAL packets against static rules — fast, simple, but not context-aware.
Default Deny	The last firewall rule: DROP ALL — block everything not explicitly permitted above.
VPN	Creates an encrypted tunnel between devices across the Internet — private, secure communication.
PPP	Point-to-Point Protocol — auth + basic encryption. Non-routable (cannot traverse networks alone).
PPTP	Tunnels PPP across networks — easy setup but weakly encrypted. Largely deprecated.
IPSec	Encrypts data using the IP framework — strong encryption, complex config, industry standard.
Router	Layer 3 device connecting separate networks — routes packets using IP addresses.
Switch	Layer 2/3 device connecting devices within a LAN — forwards frames using MAC addresses.
VLAN	Virtual LAN — logically segments a physical network on a Layer 3 switch without extra hardware.
CAM Table	A switch's MAC address table — maps MAC addresses to physical ports for efficient frame delivery.
DoS Attack	Denial of Service — floods a target with traffic to overload and crash it (demonstrated in firewall lab).

7. Security Implications

Port Forwarding Expands the Attack Surface

Every port-forwarding rule creates a publicly reachable service. Attackers continuously scan the entire Internet for open ports — automated tools like Masscan can scan the full IPv4 address space in under 6 minutes. Any forwarded port must run a hardened, patched service. Unnecessary rules must be removed immediately.

Stateless Firewalls Can Be Bypassed

Because stateless firewalls evaluate packets in isolation, a sophisticated attacker can craft packet sequences that individually appear benign but collectively represent a malicious session. Stateful firewalls prevent this by tracking the full connection context — understanding that an unexpected packet mid-session is anomalous.

VPN Security Depends on Protocol Choice

PPTP's weak encryption has known cryptographic vulnerabilities — MS-CHAPv2 authentication used by PPTP can be cracked with modern hardware. Using PPTP for sensitive communications provides a false sense of security. IPSec or modern alternatives (WireGuard, OpenVPN) must be used where data confidentiality is required.

Router Compromise = Full Network Compromise

A compromised router gives an attacker visibility into all traffic, the ability to redirect traffic (DNS hijacking), and the ability to alter firewall rules to open new entry points. Router security is therefore paramount: change default credentials immediately, disable remote management, keep firmware updated, and monitor for unauthorised configuration changes.

ARP Poisoning Targets Switches

Layer 2 switches trust ARP responses to build their MAC address tables. An attacker on the same LAN can send forged ARP replies to poison the switch's CAM table — causing traffic destined for one device to be delivered to the attacker instead (man-in-the-middle). Dynamic ARP Inspection on managed switches prevents this.

VLAN Hopping — Bypassing Network Segmentation

VLANs are a powerful segmentation tool, but misconfigured trunk ports can be exploited. VLAN hopping attacks (double-tagging) allow an attacker to inject frames into a VLAN they should not have access to, bypassing the logical separation. Strict VLAN configuration and disabling automatic trunking negotiation (DTP) are the defences.

8. Key Takeaways & Next Steps

8.1 Key Takeaways

- Port forwarding makes internal services publicly accessible — configured at the router, not the firewall.
- Firewalls are the primary perimeter defence — operating at OSI Layers 3 and 4, filtering by IP, port, and protocol.
- Stateful firewalls inspect entire connections; stateless firewalls inspect individual packets — stateful is more intelligent.
- Default-deny is the gold standard — block everything not explicitly permitted.
- VPNs create encrypted tunnels through the Internet — PPP (basic), PPTP (weak/deprecated), IPSec (strong/standard).
- Routers operate at Layer 3 and connect separate networks; switches operate at Layer 2/3 and connect devices within a network.
- Layer 3 switches support VLANs — logical network segmentation without additional physical hardware.
- Both firewall labs (DoS defence and network simulator) reinforce that correct configuration, not just presence, determines effectiveness.

8.2 Next Steps on TryHackMe

- **Immediate:** Continue with the How Websites Work room — how web traffic flows through all the networking layers covered so far.
- **Foundation:** Complete the DNS in Detail room — how domain names are resolved to IP addresses across routers.
- **Security:** Study the Network Security module — firewalls, IDS/IPS, and network monitoring in depth.
- **Offensive:** Take the Jr Penetration Tester path — apply port scanning (Nmap), firewall evasion, and network exploitation.
- **Applied:** Explore Wireshark: The Basics — see packets, frames, handshakes, and routing in live captures.
- **Certification:** Study for CompTIA Network+ for comprehensive formal networking certification.

Why This Room Matters

Every technology in this room — port forwarding, firewalls, VPNs, routers, and switches — appears constantly in both offensive and defensive security work. Penetration testers look for misconfigured port-forwarding rules and weak VPN protocols. Defenders rely on firewall rules and VLANs as their first line of control. This is not theoretical knowledge — it is the practical foundation of real network security.

Source: TryHackMe — Extending Your Network Room | tryhackme.com/room/extendingyournetwork